TASK 3: SELLER (PUBLISHER) AND INVENTORY MANAGEMENT SYSTEM

PROJECT DETAILS

- **Name:** Kaviya S
- **Roll No:** E25AI118
- **Department:** B.Sc Computer Science With AI

SCENARIO

The e-commerce bookstore (**PALETTE: Pixel & Pages**) wants to maintain information about book publishers/sellers and track physical inventory levels. The system manages seller details (such as seller ID, name, email, phone, and address) and inventory details (such as inventory ID, book ID, seller ID, stock quantity, stock status, and last updated date) to ensure stock availability, avoid stockouts, and facilitate efficient supply chain management.

1. DESIGN SELLER AND INVENTORY TABLES

i. Seller Table:

```
SQL> CREATE TABLE Seller (
  2      Seller_ID VARCHAR2(7),
  3      Seller_Name VARCHAR2(25),
  4      Email VARCHAR2(30),
  5      Phone NUMBER(10),
  6      Address VARCHAR2(45)
  7  );

Table created.
```

ii. Inventory Table:

```
SQL> CREATE TABLE Inventory (
  2      Inventory_ID VARCHAR2(7),
  3      Product_ID VARCHAR2(7),
  4      Seller_ID VARCHAR2(7),
  5      Stock_Quantity NUMBER,
  6      Stock_Status VARCHAR2(11),
  7      Last_Updated DATE
  8  );

Table created.
```

The Seller and Inventory tables have been created successfully. Now, we will define the relationships between these tables and apply necessary constraints to enforce data integrity.

2. DEFINE RELATIONSHIPS AND CONSTRAINTS

i. Relationships:

- One Seller can supply many Books/Inventory items ($1 \rightarrow *$).
- Seller_ID is the Primary Key in the Seller table and acts as a Foreign Key in the Inventory table to link the two entities.
- Inventory_ID is the Primary Key in the Inventory table.
- Book_ID acts as a Foreign Key referencing the parent Book table.

ii. Constraint Explanations:

- **PRIMARY KEY:** Uniquely identifies each row in a table (Seller_ID and Inventory_ID).
- **FOREIGN KEY:** Links the Inventory table back to the parent Book table (Book_ID) and Seller table (Seller_ID).
- **NOT NULL:** Guarantees that essential fields like Seller_Name, Email, and Stock_Quantity cannot be left empty.
- **CHECK:** Validates that Stock_Quantity is greater than or equal to zero (**≥0**) and restricts Stock_Status strictly to valid values (e.g., 'Available' or 'Out of Stock').
- **UNIQUE:** Prevents duplicate values for unique attributes like seller Email.

iii. Defining Relationship and Creating Constraints for Seller and Inventory:

a. Primary Key Constraints

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY (Seller_ID);

Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY (Inventory_ID);

Table altered.
```

b. Foreign Key Constraints

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Prod FOREIGN KEY (Product_ID) REFERENCES Book (Book_ID);

Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Seller FOREIGN KEY (Seller_ID) REFERENCES Seller (Seller_ID);

Table altered.
```

iv. Adding Constraints:

a. NOT NULL Constraints

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;

Table altered.
```

b. CHECK Constraints

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT CK_Quan CHECK (Stock_Quantity >= 0);

Table altered.
```

c. UNIQUE Constraints

```
SQL> ALTER TABLE Seller ADD CONSTRAINT Un_mail UNIQUE (Email);

Table altered.
```

3. INSERT VALUES

i. Seller Table Interactive Insert:

```
SQL> /
Enter value for seller_id: S00003
Enter value for seller_name: OReilly Media
Enter value for email: orders@oreilly.com
Enter value for phone: 7890654312
Enter value for address: 88, T. Nagar, Chennai
old   1: INSERT INTO Seller VALUES ('&Seller_ID', '&Seller_Name', '&Email', &Phone, '&Address')
new   1: INSERT INTO Seller VALUES ('S00003', 'OReilly Media', 'orders@oreilly.com', 7890654312, '88, T. Nagar, Chennai')

1 row created.

SQL> /
Enter value for seller_id: S00004
Enter value for seller_name: Rupa Publications
Enter value for email: distributor@rupa.in
Enter value for phone: 7824003009
Enter value for address: 14, Connaught Place, New Delhi
old   1: INSERT INTO Seller VALUES ('&Seller_ID', '&Seller_Name', '&Email', &Phone, '&Address')
new   1: INSERT INTO Seller VALUES ('S00004', 'Rupa Publications', 'distributor@rupa.in', 7824003009, '14, Connaught Place, New Delhi')
```

```
SQL> INSERT INTO Seller VALUES ('&Seller_ID', '&Seller_Name', '&Email', &Phone, '&Address');
Enter value for seller_id: S00001
Enter value for seller_name: Penguin India
Enter value for email: contact@penguin.in
Enter value for phone: 9876543210
Enter value for address: 12, MG Road, Bengaluru
old   1: INSERT INTO Seller VALUES ('&Seller_ID', '&Seller_Name', '&Email', &Phone, '&Address')
new   1: INSERT INTO Seller VALUES ('S00001', 'Penguin India', 'contact@penguin.in', 9876543210, '12, MG Road, Bengaluru')

1 row created.

SQL> /
Enter value for seller_id: S00002
Enter value for seller_name: HarperCollins Distro
Enter value for email: sales@harpercollins.in
Enter value for phone: 9876543012
Enter value for address: 45, Anna Salai, Chennai
old   1: INSERT INTO Seller VALUES ('&Seller_ID', '&Seller_Name', '&Email', &Phone, '&Address')
new   1: INSERT INTO Seller VALUES ('S00002', 'HarperCollins Distro', 'sales@harpercollins.in', 9876543012, '45, Anna Salai, Chennai')

1 row created.
```

ii. Inventory Table Interactive Insert:

```
SQL> INSERT INTO Inventory VALUES ('&Inventory_ID', '&Product_ID', '&Seller_ID', &Stock_Quantity, '&Stock_Status', TO_DATE('&Last_Updated', 'YYYY-MM-DD'));
Enter value for inventory_id: Inv0003
Enter value for product_id: B1003
Enter value for seller_id: S00002
Enter value for stock_quantity: 0
Enter value for stock_status: Out of Stock
Enter value for last_updated: 2026-09-15
old   1: INSERT INTO Inventory VALUES ('&Inventory_ID', '&Product_ID', '&Seller_ID', &Stock_Quantity, '&Stock_Status', TO_DATE('&Last_Updated', 'YYYY-MM-DD'
))
new   1: INSERT INTO Inventory VALUES ('Inv0003', 'B1003', 'S00002', 0, 'Out of Stock', TO_DATE('2026-09-15', 'YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for inventory_id: Inv0004
Enter value for product_id: B1004
Enter value for seller_id: S00003
Enter value for stock_quantity: 5
Enter value for stock_status: Available
Enter value for last_updated: 2026-09-18
old   1: INSERT INTO Inventory VALUES ('&Inventory_ID', '&Product_ID', '&Seller_ID', &Stock_Quantity, '&Stock_Status', TO_DATE('&Last_Updated', 'YYYY-MM-DD'
))
new   1: INSERT INTO Inventory VALUES ('Inv0004', 'B1004', 'S00003', 5, 'Available', TO_DATE('2026-09-18', 'YYYY-MM-DD'))

1 row created.
```

```
SQL> INSERT INTO Inventory VALUES ('&Inventory_ID', '&Product_ID', '&Seller_ID', &Stock_Quantity, '&Stock_Status', TO_DATE('&Last_Updated', 'YYYY-MM-DD'));
Enter value for inventory_id: Inv0001
Enter value for product_id: B1001
Enter value for seller_id: S00001
Enter value for stock_quantity: 15
Enter value for stock_status: Available
Enter value for last_updated: 2026-09-10
old   1: INSERT INTO Inventory VALUES ('&Inventory_ID', '&Product_ID', '&Seller_ID', &Stock_Quantity, '&Stock_Status', TO_DATE('&Last_Updated', 'YYYY-MM-DD'
))
new   1: INSERT INTO Inventory VALUES ('Inv0001', 'B1001', 'S00001', 15, 'Available', TO_DATE('2026-09-10', 'YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for inventory_id: Inv0002
Enter value for product_id: B1002
Enter value for seller_id: S00002
Enter value for stock_quantity: 8
Enter value for stock_status: Available
Enter value for last_updated: 2026-09-12
old   1: INSERT INTO Inventory VALUES ('&Inventory_ID', '&Product_ID', '&Seller_ID', &Stock_Quantity, '&Stock_Status', TO_DATE('&Last_Updated', 'YYYY-MM-DD'
))
new   1: INSERT INTO Inventory VALUES ('Inv0002', 'B1002', 'S00002', 8, 'Available', TO_DATE('2026-09-12', 'YYYY-MM-DD'))

1 row created.
```

iii. Display Tables:

a. Seller Table

```
SQL> SELECT * FROM Seller;

SELLER_ SELLER_NAME               EMAIL                            PHONE
------- ------------------------- -------------------------------- ----------
ADDRESS
------------------------------------------
S00001  Penguin India             contact@penguin.in               9876543210
12, MG Road, Bengaluru

S00002  HarperCollins Distro      sales@harpercollins.in           9876543012
45, Anna Salai, Chennai

S00003  OReilly Media             orders@oreilly.com               7890654312
88, T. Nagar, Chennai


SELLER_ SELLER_NAME               EMAIL                            PHONE
------- ------------------------- -------------------------------- ----------
ADDRESS
------------------------------------------
S00004  Rupa Publications         distributor@rupa.in              7824003009
14, Connaught Place, New Delhi
```

b. Inventory Table

```
SQL> SELECT * FROM Inventory;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATUS         LAST_UPDA
------- ------- ------- -------------- -------------------- ---------
Inv0001 B1001   S00001              15 Available            10-SEP-26
Inv0002 B1002   S00002               8 Available            12-SEP-26
Inv0003 B1003   S00002               0 Out of Stock         15-SEP-26
Inv0004 B1004   S00003               5 Available            18-SEP-26
```

4. GENERATE SELLER AND INVENTORY REPORTS

i. Report 1: Seller-wise Product Report

- **Display:**
    - Seller name
    - Number of products supplied
    - Available stock

```
SQL> SELECT
  2      s.Seller_Name,
  3      COUNT(i.Product_ID) AS Number_of_Products,
  4      SUM(i.Stock_Quantity) AS Available_Stock
  5  FROM Seller s
  6  JOIN Inventory i ON s.Seller_ID = i.Seller_ID
  7  WHERE i.Stock_Status = 'Available'
  8  GROUP BY s.Seller_Name;

SELLER_NAME               NUMBER_OF_PRODUCTS AVAILABLE_STOCK
------------------------- ------------------ ---------------
HarperCollins Distro                       1               8
OReilly Media                              2              10
Penguin India                              1              15
```

ii. Report 2: Stock Availability Report

- **Display:**
    - Product name
    - Stock quantity
    - Stock status

```
SQL> SELECT
  2      b.Title AS Book_Title,
  3      i.Stock_Quantity,
  4      i.Stock_Status
  5  FROM Book b
  6  JOIN Inventory i ON b.Book_ID = i.Product_ID;

BOOK_TITLE
--------------------------------------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
The Great Gatsby
            15 Available

Dune
             8 Available

Foundation
             0 Out of Stock


BOOK_TITLE
--------------------------------------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
Designing Data-Intensive Applications
             5 Available

The Great Gatsby
             5 Available
```

iii. Report 3: Inventory Analysis

- **Find:**
    - **Total products available**

```
SQL> SELECT COUNT(*) AS Total_Available_Products
  2  FROM Inventory
  3  WHERE Stock_Status = 'Available';

TOTAL_AVAILABLE_PRODUCTS
------------------------
                       4
```

    - **Products out of stock**

```
SQL> SELECT COUNT(*) AS OutOfStock_Products
  2  FROM Inventory
  3  WHERE Stock_Status = 'Out of Stock';

OUTOFSTOCK_PRODUCTS
-------------------
                  1
```

- **Highest stocked products**

```
SQL> SELECT b.Title AS Book_Title, i.Stock_Quantity
  2  FROM Book b
  3  JOIN Inventory i ON b.Book_ID = i.Product_ID
  4  WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);

BOOK_TITLE
--------------------------------------------------------------------------------
STOCK_QUANTITY
--------------
The Great Gatsby
            15
```

- **Average inventory quantity**

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory
  2  FROM Inventory;

AVERAGE_INVENTORY
-----------------
              6.6
```